

DMBI Unique Questions

Module 1: Data Mining & KDD Process

#	Question	Appeared In
1	Define data mining. Explain the KDD process with a suitable diagram.	DEC 23, DEC 24, MAY 23
2	List stages in Data Mining with a diagram.	MAY 24
3	What is a decision support system?	MAY 25

Module 2: Data Warehouse, OLAP & Schemas

#	Question	Appeared In
1	Define data warehouse. Explain data warehouse architecture with a diagram.	DEC 23
2	Draw components of data warehouse architecture.	MAY 24
3	Draw Data warehousing Architecture.	MAY 23
4	Give the difference between OLAP and OLTP.	DEC 24
5	Compare OLTP and OLAP.	MAY 23
6	What is OLAP? Explain OLAP operations (Roll-up, Drill-down, Slice, Dice, Pivot) with diagrams/examples.	MAY 24, MAY 25
7	Compare star schema and snowflake schema.	DEC 24
8	Compare star schema, snowflake schema, star constellation.	MAY 23
9	Explain Star vs Snowflake schema with example.	MAY 24
10	Dimensional Modeling.	MAY 23
11	BI Architectures.	MAY 24

Module 3: Data Pre-processing & Attributes

#	Question	Appeared In
1	Explain types of attributes used in data exploration.	DEC 23
2	Explain different types of attributes used in data exploration with examples.	DEC 24
3	Types of attributes.	MAY 24
4	What is noisy data? How to handle it?	DEC 23, DEC 24, MAY 23
5	Steps in data preprocessing.	DEC 24
6	What is an outlier? List types of outliers. Describe methods used for outlier analysis.	DEC 23
7	What is an outlier? Types and methods for outlier analysis.	DEC 24, MAY 23, MAY 24
8	Two outlier detection methods.	MAY 25

Module 4: Clustering & Association Rule Mining

#	Question	Appeared In
1	Explain DBSCAN algorithm with example.	DEC 23, DEC 24, MAY 23
2	Explain K-means algorithm in detail.	DEC 23, DEC 24
3	What is K-Means clustering?	MAY 25
4	Steps in hierarchical clustering.	MAY 24
5	What is BIRCH algorithm?	MAY 24, MAY 25
6	Explain Multilevel and Multidimensional Association rules with suitable examples.	DEC 23, DEC 24
7	Explain Market Basket Analysis with example.	MAY 23
8	What is market basket analysis? Real use case.	MAY 24
9	Explain mining of Multilevel and Multidimensional association rules.	MAY 24
10	How to improve efficiency of Apriori algorithm?	MAY 25

Module 5: Classification & Regression

#	Question	Appeared In
1	Difference between classification and prediction with examples.	MAY 25
2	Explain decision tree classifier. Steps for ID3 algorithm with example.	MAY 24
3	Decision Tree Induction.	MAY 23
4	Explain information gain and Gini value used in decision tree.	MAY 23
5	Attribute selection in decision trees. Information Gain and Gini Index with formulas.	MAY 25
6	Naïve Bayes Classifier.	MAY 24
7	Short note on simple linear regression.	MAY 25

Module 6: Ensemble Methods, BI & Evaluation

#	Question	Appeared In
1	Compare Bagging and Boosting of a classifier.	DEC 23
2	Explain Bagging of a classifier.	DEC 24
3	What are ensemble methods? Why useful?	MAY 25
4	Random Forest.	MAY 23
5	Boot Strapping.	MAY 24
6	Cross Validation.	MAY 23
7	Explain Confusion Matrix.	DEC 24
8	Design a BI system for Fraud Detection. Explain all steps from data collection to decision making.	DEC 23, DEC 24
9	Differences between Business Intelligence and traditional reporting systems.	MAY 25